DATABRICKS · UPSTREAM SUBSURFACE

Subsurface Intelligence

From raw well logs to a located, valued, governed drilling decision, on one platform, in one sitting.

$40M
risked NPV10 of the next well, found in data the team already has

3.6x
lower error filling a missing sonic log than today's method

1 sitting
from raw logs to a located, priced, governed well

Locate → Read → Fill the sonic → Model in voxels → Value → Decide

Working demo on Databricks · Lakeflow · Unity Catalog · MLflow · Model Serving · AI Gateway · Spatial SQL · Genie · Apps

01 · THE DECISION

Where does the next well go, and is it worth drilling? The answer is spread across five systems.

DATA IN SILOS

Logs, tops, OSDU records, economics, maps

Each lives with a different team and tool. Every new question starts with an export and a hand-off.

GAPS IN THE RECORD

Wells drilled without a key curve

A missing sonic log blocks geomechanics and seismic ties. The usual fix is an empirical relation applied by hand.

AI WITHOUT GUARDRAILS

Assistants that cannot see, or ignore, entitlements

Either the model has no access to the governed data, or it has too much and partners see what they should not.

Subsurface Intelligence puts the logs, the geomodel, the economics and the AI on one governed platform, so the decision is one workflow instead of five projects.

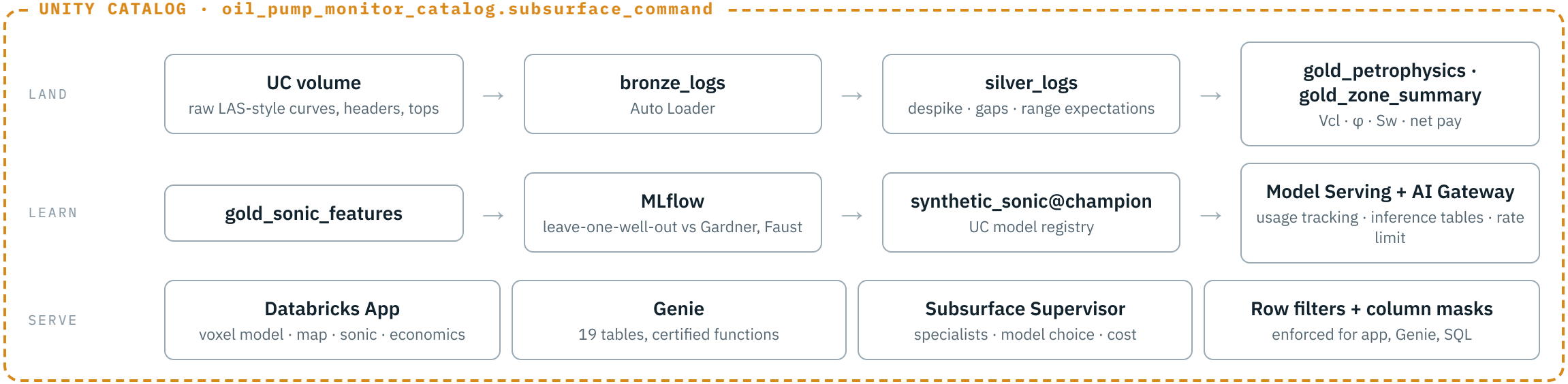
Four levers: a better well, found sooner, without blocked decisions, safe to share

LEVER	WHAT CHANGES	THE NUMBER
1. Value per decision	The geomodel finds and ranks a step-out in data the team already has, and the certified NPV function prices it	**$103M** NPV10 unrisked, **$40M** risked, 34% IRR, $53 break-even on $58M capex demo
2. Speed to decision	Logs to a located, priced proposal in one sitting instead of a chain of exports and hand-offs	Every **6 months** of delay costs **$1.9M** of the risked value (**$4.8M** unrisked), 4.7% of NPV at 10% demo. How long does a step-out proposal take today? yours
3. No blocked decisions	Missing curves are filled in seconds by a served model scored against the method engineers trust	**3.6x** lower error than calibrated Gardner on 13 held-out wells demo. How many of your wells lack sonic? yours
4. Safe to share	Partners and AI assistants see only what they are entitled to, enforced once in Unity Catalog	JV partner sees **30 of 84** rows, interpretation masked, in the app, Genie and SQL demo

The cost side stays small: the sonic model is a scale-to-zero endpoint metered through AI Gateway, the pipeline runs on serverless, and the AI shows its cost per run. Value per decision is in the tens of millions; the platform cost per decision is not.

03 · HOW IT IS BUILT

One pipeline, one registry, one governance layer



One Asset Bundle job, **subsurface-command-build**, lands the data, runs the pipeline, trains, registers, serves and scores. Rebuilding the whole stack is one command. Cost stays visible: serverless pipeline, a scale-to-zero endpoint with usage tracking and inference tables, and per-run token cost on the Supervisor.

04 · THE PATH, AND THE ASK

Prove it on one field with your logs, then reuse the template

WEEKS 1–2

Land

Your LAS files, tops and headers for one field into a UC volume; same pipeline, your data.

WEEKS 3–4

Score

Retrain the sonic model on your wells; agree the bar it must clear against your current method.

WEEKS 5–8

Decide

Voxel model on your horizons; one step-out reviewed by your geoscientists against their own pick.

AFTER

Scale

Next field, next missing curve, partner access under the same policies.

ASK 1 · BUSINESS

Name one field and the person who owns its next-well decision

Success is a step-out your team agrees is credible, with its value, in weeks instead of a quarter.

ASK 2 · TECHNICAL

A two-week working session with petrophysics and data

Map your logs, tops and OSDU records to the pipeline, and agree the acceptance bar for the sonic model.

APPENDIX · WHAT IS REAL, AND WHAT IS ILLUSTRATIVE

Every screen says where its numbers come from

ELEMENT	STATUS	DETAIL
Pipeline, model, endpoint, policies, Genie	Live	Running on Databricks; job, endpoint and policies verified
Sonic predictions	Live	Served model on every request; labelled Gardner fallback if cold
Voxel properties and step-out	Live	Computed in the app from the wells' logs and tops on every load
Well logs	Synthetic	Physically coupled rock model (porosity, shale, fluids); your logs replace them in the pilot
Structural framework	Illustrative	Dip, closure and fault are modeled, honoured at the tops; a real horizon model plugs in
OSDU analog panels	Offline	Fed by a partner ADME workspace that is not live; the panels say so